

Experiment Manual: Evaluation of Sedimentation Efficiency and Settleable Solids in Wastewater

Objectives

- Understand sedimentation as a primary wastewater treatment process.
- Measure settleable solids using a graduated cylinder.
- Evaluate clarification using UV–Vis absorbance (600 nm) and/or TSS/turbidity.
- Determine optimum settling time and removal efficiency.

Theory

Sedimentation is a gravity separation process in which suspended particles settle when their density exceeds that of water. It is the principal operation in primary clarifiers and reduces the load on downstream biological treatment. Larger and denser particles settle faster than fine or colloidal particles. This laboratory is a bench-scale process evaluation, not the APHA Imhoff cone test. Supernatant clarity is evaluated using UV–Visible spectrophotometry at 600 nm (an indirect indicator of turbidity), and optionally by TSS or a nephelometer.

Apparatus

- 1000 mL graduated cylinder
- 1000 mL beaker
- Pipette
- Stopwatch
- Ruler
- UV–Vis spectrophotometer
- Glass cuvettes
- Filter apparatus (optional)

Sample

Raw domestic wastewater, composite tannery wastewater, or synthetic wastewater (5 g fine clay in 1 L water). Mix thoroughly before use.

Procedure

1. Measure initial UV–Vis absorbance at 600 nm using distilled water as blank. Optionally measure TSS/turbidity.
2. Transfer 1000 mL sample to graduated cylinder and start timer.
3. Allow settling undisturbed.
4. At 10, 20, 30, 45 and 60 min record sludge height/volume.
5. At each interval withdraw ~10 mL supernatant from ~5 cm below surface without disturbing sludge.
6. Measure absorbance at 600 nm. Optionally determine TSS/turbidity.

7. Repeat in duplicate if possible.

Observation Table

Time(min)	Sludge height(cm)	Sludge vol.(mL/L)	Abs600	Turbidity	TSS	Remarks
0						
10						
20						
30						
45						
60						

Calculations

Settleable solids (mL/L)=Observed sludge volume (mL) for a 1 L sample.

Removal (%) = ((Initial – Final)/Initial) ×100.

For UV–Vis: Clarification (%)=(($A_0 - A_t$)/ A_0)×100.

For TSS: Removal (%)=(($S_0 - S_t$)/ S_0)×100.

Graphs

- Sludge volume vs time
- Absorbance vs time
- TSS/turbidity vs time
- Removal efficiency vs time

Expected Results

Sludge volume increases then plateaus. Absorbance, turbidity and TSS decrease with settling time. Most removal occurs within the first 30–45 min.

Precautions

- Mix sample before test.
- Avoid vibration.
- Withdraw supernatant gently.
- Use clean cuvettes.
- Blank UV–Vis with distilled water.
- Do not disturb sludge layer.